

Enhancing recipe retrieval with foundation models

F. Song, B. Zhu, Y. Hao, and S. Wang, "Enhancing Recipe Retrieval with Foundation Models: A Data Augmentation Perspective," *arXiv:2312.04763 [cs.IR]*, Dec. 2023. doi: 10.48550/arXiv.2312.04763.

Enhancing with Foundation Models

The Problem

You have a food photo. You want the recipe and vice versa

Unuseful information:

Preheat oven,
plates, knives



"The cooking process is invisible in the image"

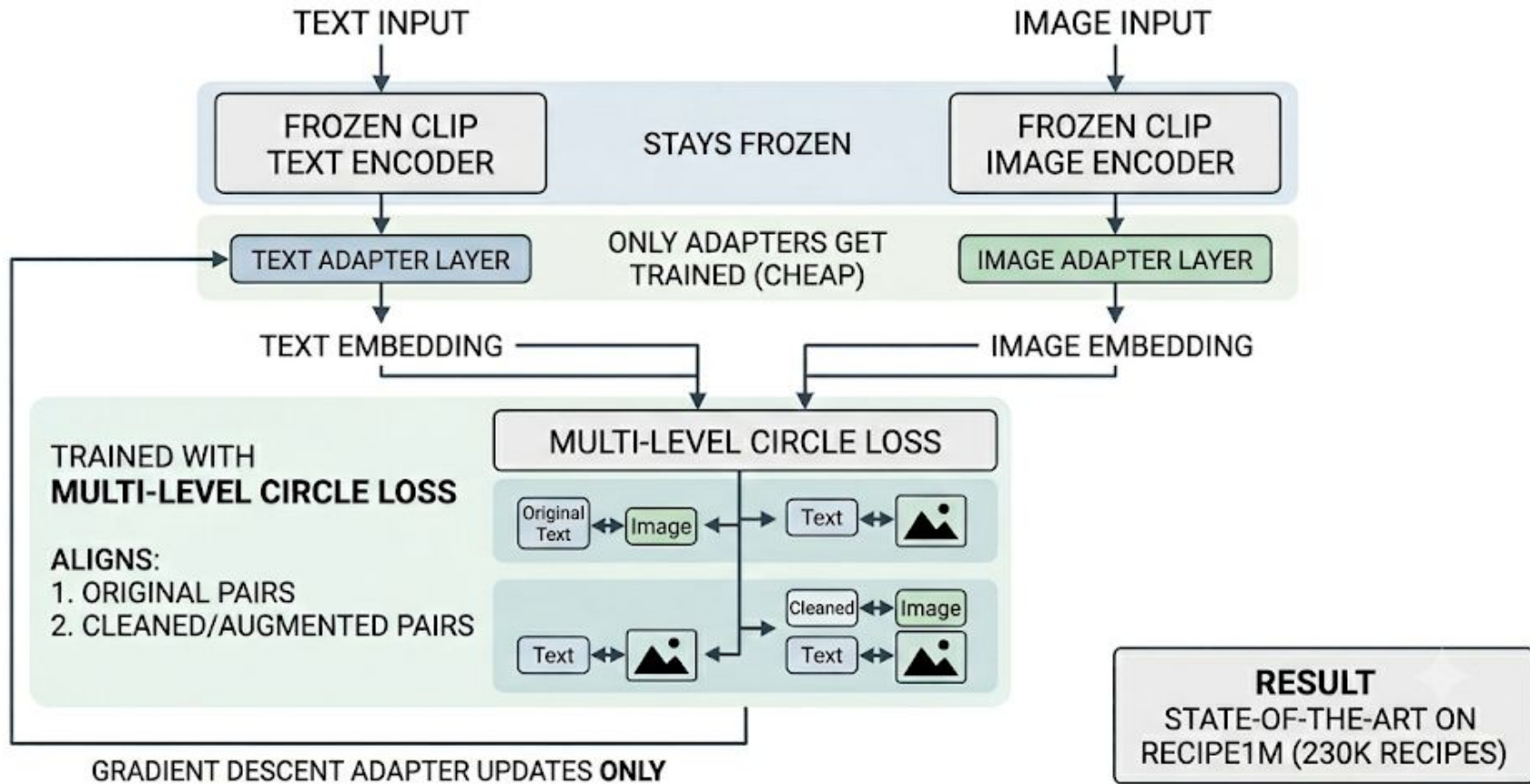
Foundation Models as Filters

Instead of training a smarter model, the paper uses two powerful pre-trained models to clean the data first:

What's noisy	Foundation model used	What it produces
Recipe text (too long, has irrelevant steps)	Llama 2 (LLM)	A 30-word "visual imagination" — e.g. "golden roasted chicken with crispy skin, surrounded by herbs"
Food image (has background, plate, tablecloth)	SAM (Segment Anything Model)	Cropped segments of just the food ingredients

Think of it like this: Llama2 reads a recipe and imagines what the dish looks like. SAM looks at a photo and isolates the food.

THE DAR ARCHITECTURE



Architecture Comparison

	DAR (paper)	Our solution
Base model	CLIP ViT-B/16 (frozen)	CLIP ViT-L/14 (frozen)
Adapter layers	Yes - down/up projection	Yes - 2-layer MLP with residual
Loss function	Multi-level circle loss	InfoNCE contrastive loss
Data augmentation	Llama2 + SAM	None (yet)
Dataset	Recipe1M (238k pairs)	Food.com + Food101 (5k pairs)

What the paper validates for us



CLIP + lightweight adapters is the right approach

Better than fully fine-tuning CLIP and significantly more cost-effective.



Data augmentation step is essential

Critical to solve misalignments: Recipe instruction $\neq$ visual representation of food.

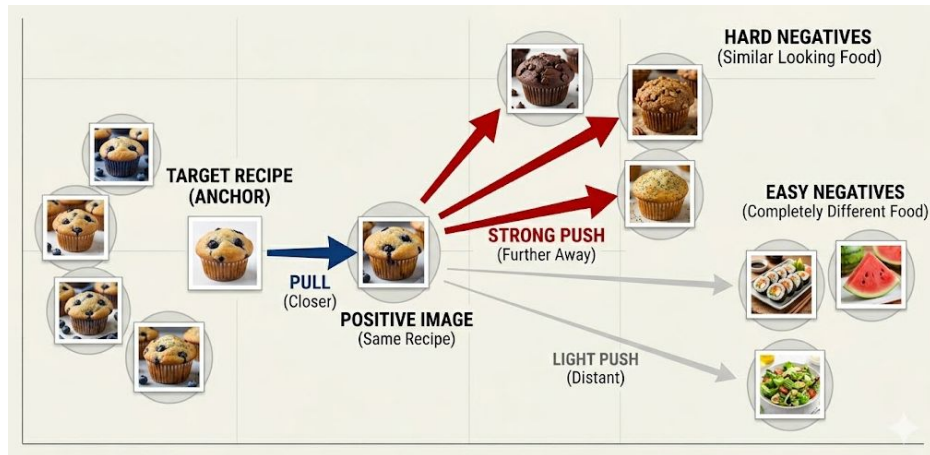
Why Multi-Level Circle Loss?

The Problem: Standard InfoNCE

Standard contrastive loss penalizes all wrong matches equally. It treats mistaking a "blueberry muffin" for a "chocolate chip muffin" the same as mistaking it for a "steak".

The Solution: Circle Loss

Dynamically re-weights penalties by aggressively pushing away "hard negatives" (visually similar incorrect dishes) to fine-tune decision boundaries.

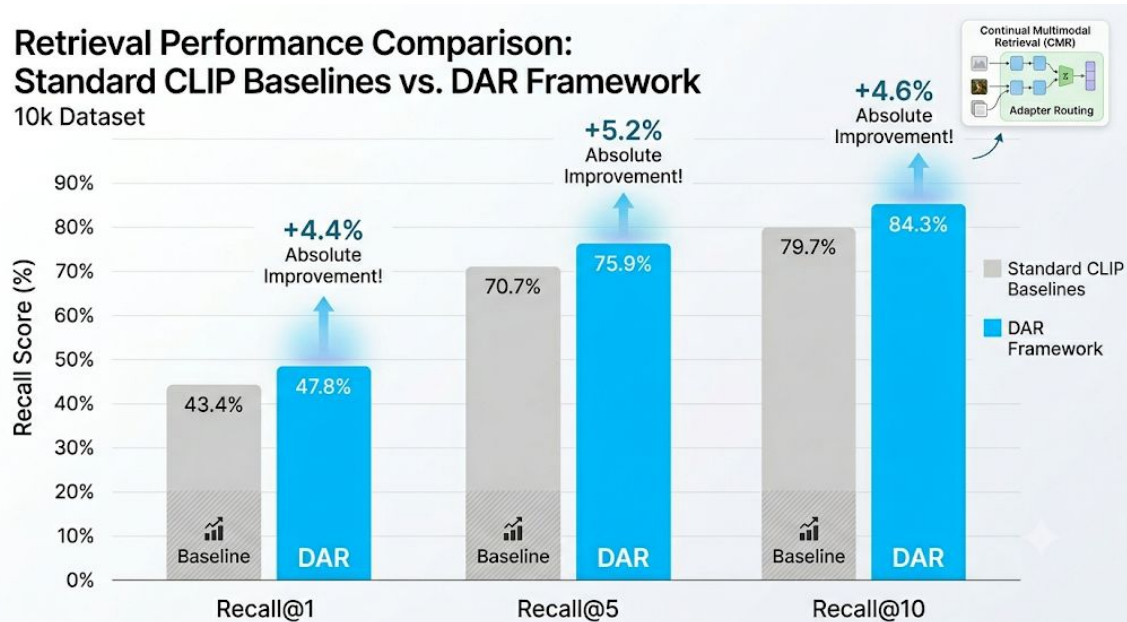


The "Multi-Level" Advantage

Calculates loss across noisy original data and clean SAM/LLM data simultaneously for maximum alignment.

Results of DAR Framework

Retrieval Performance Comparison:
Standard CLIP Baselines vs. DAR Framework
10k Dataset



Setting the Benchmark

Achieved unprecedented Recall@K scores on Recipe1M, outperforming larger, expensive models.

The Key Insight

Data cleaning (SAM/LLM) and loss optimization are more effective than scaling parameters.

Efficiency

Uses lightweight adapters, keeping computational costs extremely low.

Project Roadmap

Phase 1: The Data Pipeline



Building the foundation for high-quality training data.

- Integrate local LLM to rewrite Kaggle instructions into visual descriptions.
- Apply SAM to Food101 images to remove background noise (plates, tables).

Phase 2: The Objective Upgrade



Optimizing learning signals for complex food hierarchies.

- Swap out InfoNCE loss for Multi-Level Circle Loss.
- Enable fine-grained visual-text alignment.

Phase 3: Scaling Up



Full-scale deployment of the automated pipeline.

- Scale training to full 150k+ dataset.
- Confidently process noisy data without model confusion.

Thank you for your attention!